Echoes and Delays: Time-to-Build in Production Networks

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Motivation

- Firms in a modern economy rely on a complex network of suppliers, whose time-to-build varies significantly
  - Ex.: $<$ 1 month for furniture assembly, 1-2 years for container ships

- Most of production network literature ignores time-to-build and delays
  - Acemoglu et al. (2012), Baqaee and Farhi (2019, 2020), ... essentially static
  - Roundabout production: inputs are produced and used in the same period, so disruptions are resolved within the period

- How does the introduction of time-to-build or delivery lags affect the dynamics of production networks?

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- A simple model of time-to-build (T2B) in a production network
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- We analyze how T2B shapes the propagation of shocks:
  1. Persistence and bottlenecks
     $\Rightarrow$ shocks take longer to propagate through the economy
  2. Echoes: cycles in the network make a shock come back
     $\Rightarrow$ cyclical fluctuations
  3. Aggregate fluctuations
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- We test the model's timing predictions in monthly U.S. industrial production data
  - customers move with their suppliers at the measured delays
  - aggregate implications visible in the data

Related Literature

- Shock propagation in production networks
  - Acemoglu et al. (2012), Barrot and Sauvagnat (2016), Acemoglu et al. (2017), Baqaee and Farhi (2019), Carvalho et al. (2021), Ghassibe (2024), ...
- Time-to-build
  - Kydland and Prescott (1982), Schwartzman (2014), ...
  - In production networks: Long and Plosser (1983), Carvalho and Reischer (2021), Liu and Tsyvinski (2024), Bizzarri (2024), Bizzarri, Pangallo and Queirós (2025), Leng et al. (2025)
- Endogenous fluctuations in multi-sector economies
  - Benhabib and Nishimura (1979, 1985, ...)
- Delays in supply chains
  - Djankov et al. (2010), Hummels and Schaur (2013), Alessandria et al. (2023), Carreras-Valle and Ferrari (2025)

More

Data

Input-output network

- BEA use table for 2017: 402 industries, 371 sectors kept

Measuring delays

$$\text{backlog ratio} = \frac{\text{stock value of unfilled orders}}{\text{flow value of goods delivered per month}}$$

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1. Census M3 survey: mostly durable-goods ind., 124 sectors
2. Compustat: listed firms, many outside manufacturing: 112 sectors
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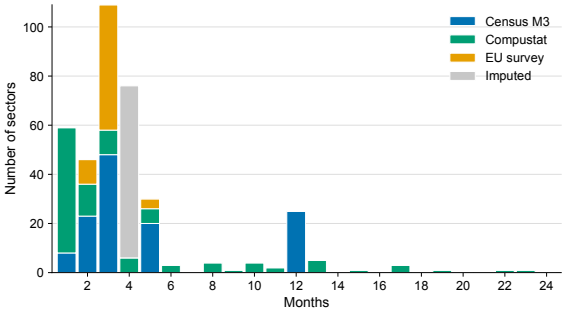
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The 70 other sectors (agriculture, services) get the measured manufacturing average, 4 months

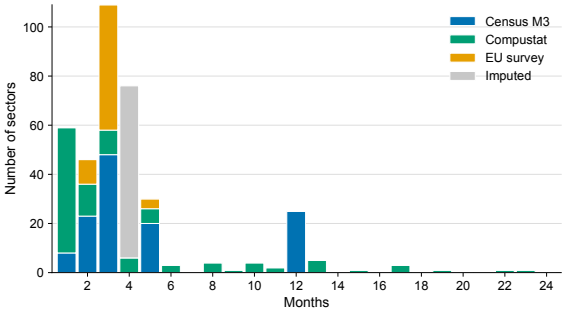
Intuition Data sources

Backlog Distribution



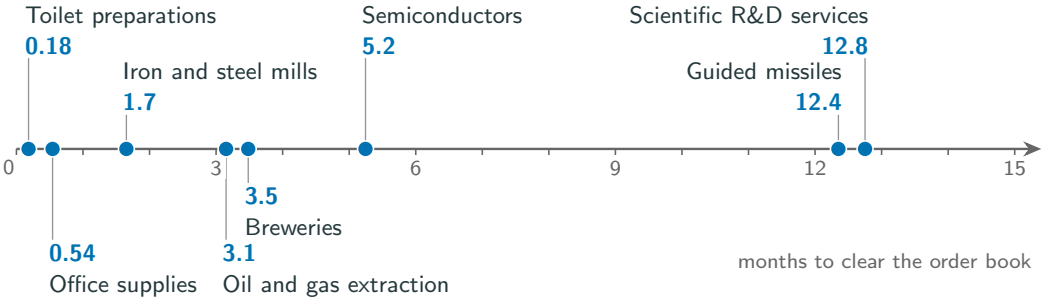
- Average 4.2 months

Backlog Distribution



- Average 4.2 months
- Measured for 301 sectors (62.6% of value added)
- Imputed for the rest (mostly agriculture and services)
- Most results insensitive to imputation (IP)

Delays range from days to more than a year



Backlog ratios in months, before rounding

The whole network

Model

Model

- Discrete time
- Representative household with inelastic labor supply, discount factor β
- Sectors $n = 1, \ldots, N$ with production

$$y_{nt} = A_{nt} F_n \left(l_{nt}, x_{n1,t}, \ldots, x_{nN,t} \right)$$

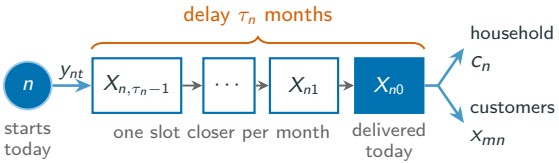
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  where A_{nt} is productivity, l_{nt} labor, $x_{nm,t}$ input of good m
- Time-to-build modeled as delivery lags:
  - Goods of sector n take τ_n periods to be delivered (the *delay* of n)
  - $X_{ns} \equiv$ stock of good n scheduled for delivery in s periods



No capital, no storage between periods

Planning Problem

- Complete markets $\Rightarrow$ planning problem

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$$V\left(\{A_n\}, \{X_{1s}\}_{s=0}^{\tau_1-1}, \ldots, \{X_{Ns}\}_{s=0}^{\tau_N-1}\right) = \max_{c_n, l_n, x_{nm}, y_n} U\left(c_1, \ldots, c_N\right) + \beta E\left[V\left(\{A_n'\}, \{X_{1s}'\}_{s=0}^{\tau_1-1}, \ldots, \{X_{Ns}'\}_{s=0}^{\tau_N-1}\right)\right]$$

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subject to the labor constraint $\sum_n l_n \leq 1$ and, for all $n = 1, \ldots, N$:

$X'_{ns} = X_{n,s+1}$ for $0 \leq s < \tau_n - 1$	goods in the pipeline move one period closer
$X'_{n,\tau_n-1} = y_n$	output started today is delivered in τ_n periods
$X_{n0} \geq c_n + \sum_m x_{mn}$	goods delivered today are consumed or used as inputs
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$\Rightarrow$ The whole pipeline is in the state, 371 sectors $\times$ their lags: intractable in general

Solution

- Way out: log utility and Cobb-Douglas production with constant returns

$$U = \log\left(\prod_n c_n^{\gamma_n}\right), \qquad F_n\left(I, x_1, \ldots, x_N\right) = I^{\alpha_n} \prod_m x_m^{\omega_{nm}}, \quad \alpha_n + \sum_m \omega_{nm} = 1$$

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Proposition

1. *Policy functions are* $l_n = \bar{l}_n$, $c_n = \bar{c}_n X_{n0}$, $x_{mn} = \bar{x}_{mn} X_{n0}$ *(bar variables are constant),*
2. *The value function is*

$$V = \sum_{n=1}^{N}\sum_{s=0}^{\tau_n - 1} \beta^s \zeta_n \log X_{ns} + G\left(\{A_n\}\right),$$

with ζ_n *and* G *given on the next slide.*

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- V = value of the goods in the pipeline (ζ) + value of productivity (G)

What ζ_n and G capture

- ζ_n = value of one more percent of good n, *once delivered*:

$$\zeta_n = \underbrace{\gamma_n}_{\substack{\text{consumption} \\ \text{of } n}} + \sum_m \underbrace{\omega_{mn}}_{\substack{\text{output of customer } m \\ \text{rises by } \omega_{mn}\%}} \underbrace{\beta^{\tau_m}}_{\substack{\text{delivered} \\ \tau_m \text{ periods later}}} \underbrace{\zeta_m}_{\substack{\text{worth } \zeta_m \\ \text{once delivered}}}$$

  - Iterating: ζ_n adds up every supply chain from n to the household
  - Without delays, $\boldsymbol{\zeta} = (\mathbf{I} - \boldsymbol{\Omega}')^{-1}\boldsymbol{\gamma}$: the Domar weights of the static network

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  - With i.i.d. shocks the last term is a constant: a 1% shock to n is worth $\beta^{\tau_n}\zeta_n$
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- Decentralization: prices, GDP, and Hulten's theorem

Prices GDP and Hulten

Output

- In log-deviation from steady state:

$$\hat{y}_{nt} = \hat{A}_{nt} + \sum_m \omega_{nm} \hat{y}_{m,t-\tau_m}$$

each input enters with its supplier's delay τ_m

- VAR(τ_{max}) representation:

$$\hat{\mathbf{y}}_t = \mathbf{\Omega}_1 \hat{\mathbf{y}}_{t-1} + \ldots + \mathbf{\Omega}_{\tau_{max}} \hat{\mathbf{y}}_{t-\tau_{max}} + \hat{\mathbf{A}}_t, \qquad \mathbf{\Omega}_\tau = \mathbf{\Omega}\, \mathbf{diag}\left(\mathbf{1}\{\tau = \tau_m\}\right)$$

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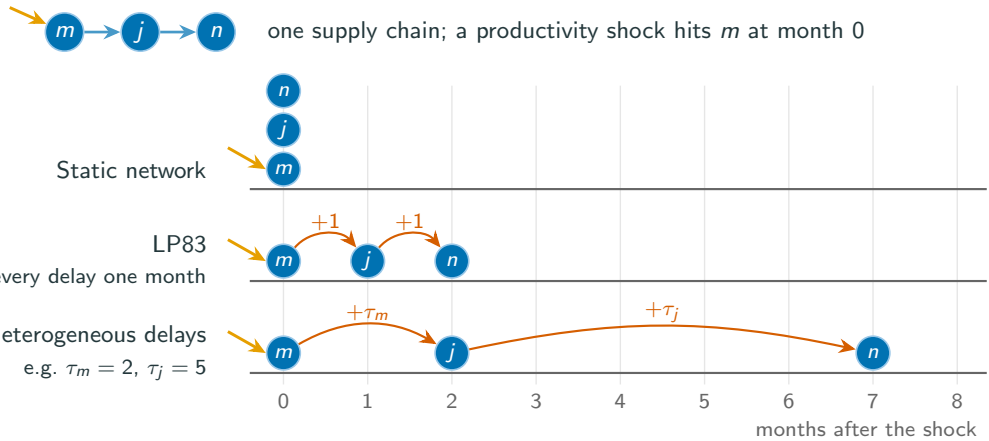
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- Nested cases:
  - Roundabout production (static network): no time-to-build

$$\hat{\mathbf{y}}_t = \hat{\mathbf{A}}_t + \mathbf{\Omega}\hat{\mathbf{y}}_t \;\Rightarrow\; \hat{\mathbf{y}}_t = \hat{\mathbf{A}}_t + \mathbf{\Omega}\hat{\mathbf{A}}_t + \mathbf{\Omega}^2\hat{\mathbf{A}}_t + \ldots = [\mathbf{I} - \mathbf{\Omega}]^{-1}\hat{\mathbf{A}}_t \quad \text{(Leontief inverse)}$$

  - Long and Plosser (1983), *LP83*: one-period time-to-build

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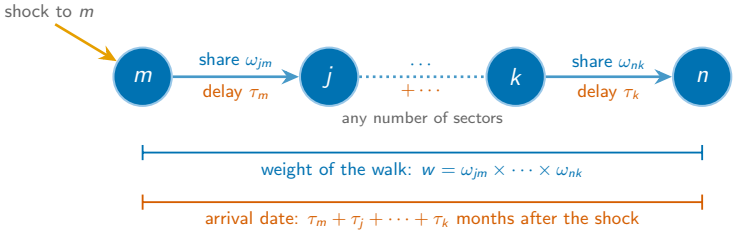
Same shock, three timings



⇒ Same chain in all three economies; the delays set only when each sector responds

Propagation along walks

- *Walk*: a supply chain of any length, $m \to j \to \cdots \to k \to n$
  - its *weight*: the product of the input shares along it
  - its duration: the sum of the delays along it, the last sector excepted



- *Cycle*: a walk from sector m to itself
  - its *weight*: same
  - its *duration*: the time it takes to come back

Persistence and Delays

Persistence and Delays: an overview

- Average duration of a shock: the mean date at which its effect reaches the household
  - ⇒ agg. shock: 7.8 months vs 2.0 in LP83

Duration Common delay

- Welfare cost: gains that arrive later are discounted
  - ⇒ the delays cost 2.5% of the static value of productivity gains vs 0.6% in LP83

Welfare

- Bottlenecks: which sectors' delays matter most for persistence?
  - ⇒ Bottlenecks identified by (weighted) supplier × buyer centrality measures

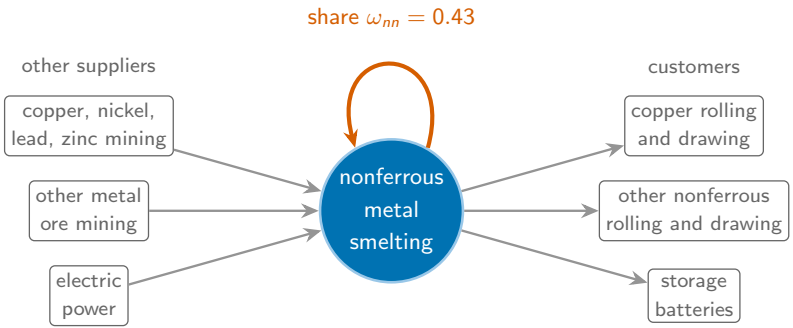
Bottlenecks

⇒ Delays make shocks slow to reach the household, unevenly across sectors

Echoes

An example: nonferrous metal smelting uses its own output

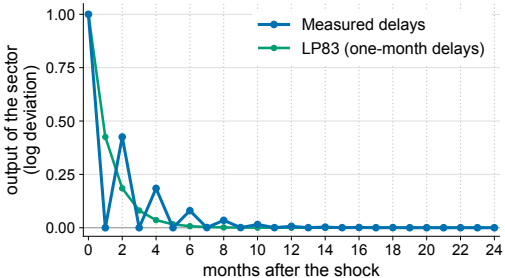
Sector 47, Nonferrous metal smelting, as a delay of $\tau_n = 2$ months



Cycle properties

- Duration = sum of the delays $\Rightarrow \tau_n = 2$ here
- Weight = product of input shares $\Rightarrow \omega_{nn} = 0.43$ here

Nonferrous metal smelting echoes every two months



Output after a shock to the sector's own productivity

- Period: 2 months = the duration of the self-loop
- Decline: each wave is 43% of the previous one = the weight of the self-loop
- LP83: smooth decline, same total

Other sectors

- Why: the shock moves output today; the sector buys that output from itself, so its inputs move two months later, 43% as much, then its output, then its inputs again, and so on

$\Rightarrow$ Period = duration of the cycle. Decline = its weight. General property?

Does the whole economy behave like this sector? The eigenvalues

- Write the VAR(τ_{max}) as a VAR(1) by stacking the lags:

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- Response t months after a shock $\mathbf{e}_0$: $\mathbf{Y}_t = \mathbf{O}^t \mathbf{e}_0$

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- Each application of $\mathbf{O}$ multiplies the j-th piece by its eigenvalue ξ_j:

$$\mathbf{Y}_t = \mathbf{O}^t \mathbf{e}_0 = \sum_j \xi_j^t \, a_j \mathbf{v}_j$$

- Time dependance only through eigenvalue ξ_j^t; the shock only sets a_j

If $\mathbf{O}$ is not diagonalizable

What one eigenvalue does to the response

Key object: ξ_j^t

- Polar form $\xi = |\xi| e^{i\theta}$, so $\xi^t = |\xi|^t e^{i\theta t}$
- Modulus $|\xi| < 1$: the term shrinks by $|\xi|$ every month, how fast it dies out
- Argument θ: whether the term oscillates, and at which period
  - $\theta = 0$ (positive real): smooth decay, no oscillation
  - $\theta = \pi$ (negative real): the sign flips every month, a period of 2 months
  - complex pair: a real oscillation with a period of $2\pi/\theta$ months

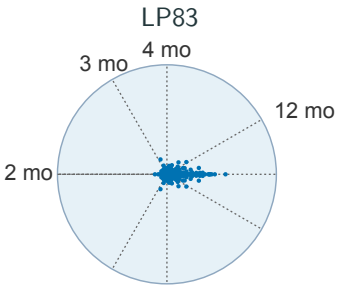
See it

- The economy oscillates when $\mathbf{O}$ has complex eigenvalues
  - Static network: all eigenvalues are zero, no dynamics
  - LP83: $\mathbf{O} = \mathbf{\Omega}$

The eigenvalues of O

Each dot: one eigenvalue. Distance to the center = modulus; rays = periods $2\pi/\theta$.

Histogram

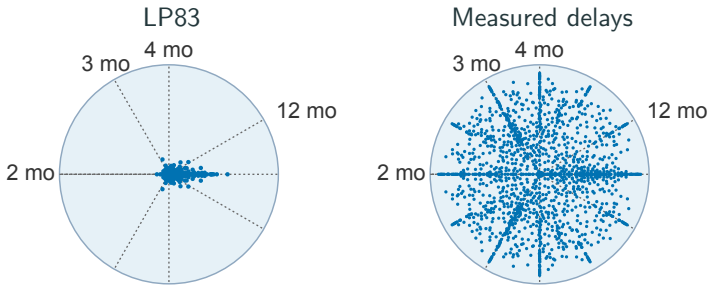


- LP83: small, near the positive real axis: shocks die out quickly and smoothly

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Histogram



- LP83: small, near the positive real axis: shocks die out quickly and smoothly
- Measured delays: larger (up to 0.92), along rays at periods of 2, 3, 4 and 12 months

$\Rightarrow$ Same input-output matrix in both. Where do the oscillations come from?

Where the eigenvalues come from: one cycle case

At $t = 0$, shock to a sector n that belongs to a single cycle of weight w and duration τ

- At $t > 0$, n moves only when the shock comes back through the cycle τ months later

$$\hat{y}_{n,t} = w\, \hat{y}_{n,t-\tau} \qquad (t > 0)$$

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- $k = 0$: smooth decay; $k = 1$: period τ (fundamental); $k \geq 2$: periods τ/k (harmonics)

Why the harmonics Poisson delays

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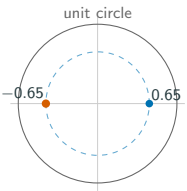
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$\Rightarrow$ Period = the duration, decay = the weight

Where the eigenvalues come from: two self-loops of the U.S. network

Nonferrous metal smelting

$w = 0.43$, $\tau = 2$

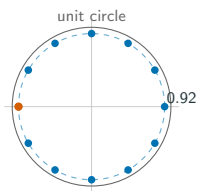


radius $0.43^{1/2} = 0.65$

- two roots, ± 0.65
- -0.65: the two-month echo

Missile propulsion units

$w = 0.37$, $\tau = 12$

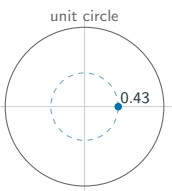


radius $0.37^{1/12} = 0.92$

- twelve roots of mod. 0.92
- 12, 6, 4, 3, 2.4, 2 months

Nonferrous metal under LP83

$w = 0.43$, $\tau = 1$



radius $0.43^{1/1} = 0.43$

- one root, 0.43
- no oscillation

Where the eigenvalues come from: any network

- In general n's output comes back to it along many routes: the walks ς from n back to n,
- After the $t = 0$ shock, n moves only when an earlier movement returns:

$$\hat{y}_{n,t} = \sum_{\varsigma} w(\varsigma)\, \hat{y}_{n,t-\tau(\varsigma)}$$

- Same as before: ξ^t reproduces itself if

$$R_n(\xi^{-1}) \equiv \sum_{\varsigma} w(\varsigma)\, \xi^{-\tau(\varsigma)} = 1$$

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1. *No network cycle* $\Rightarrow$ *every eigenvalue of* $\mathbf{O}$ *is zero: no oscillation*
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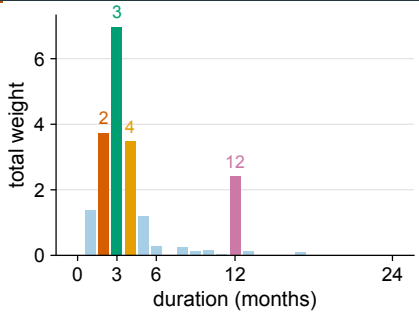
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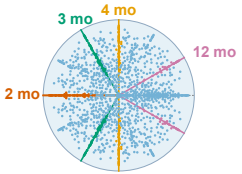
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$\Rightarrow$ Time series oscillations come from network cycles Why (iii)

The durations of the heavy cycles are the rays of the eigenvalues



Total cycle weight by duration (7,414 cycles above 10^{-6})



Eigenvalues of $\mathbf{O}$, measured delays

- Weight at 2, 3, 4 and 12 months: eigenvalues on the rays $2\pi k/\tau$ (same colors)
- Mostly self-loops (97% of what returns to a sector); heaviest two-sector cycle 0.025
- Caveat: the 4-month weight is sectors without backlog data, set to the default (98% of it)

$\Rightarrow$ The cycles set the periods at which the economy oscillates

Census Self-loops LP83

From eigenvalues to autocorrelations

Eigenvalues

- Give frequencies of oscillations, not how much each sector oscillates
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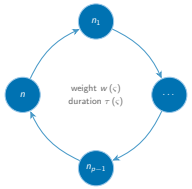
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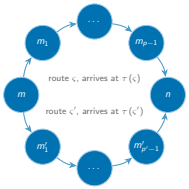
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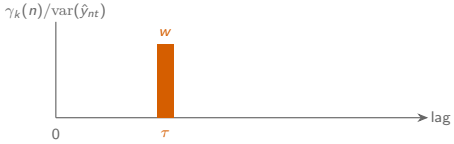
(a) Directed cycle: same route, extra laps



(b) Undirected cycle: two routes

A cycle leaves a comb in the autocorrelations





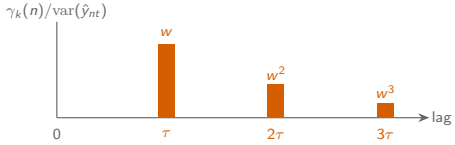
Proposition (contribution of the cycles)

Shocks i.i.d. across months and sectors; sector n lies on directed cycles c with weights w_c and durations τ_c. For every lag k,

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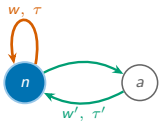


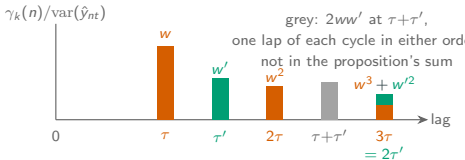
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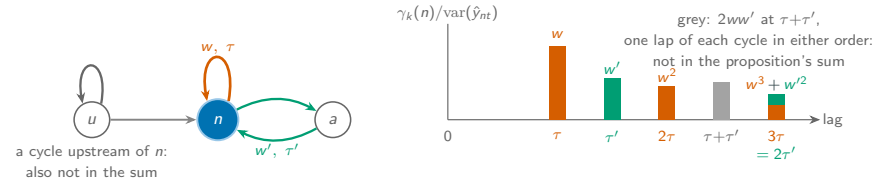


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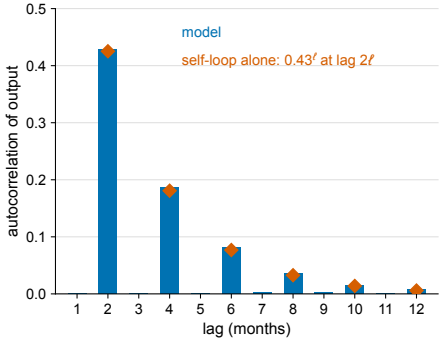


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The comb in the model: nonferrous metal



Nonferrous metal has a self-loop of weight 0.43 and duration 2.

- Bars: output autocorrelation. Diamonds: 0.43^ℓ at lag 2ℓ, the contribution of the self-loop.
- The self-loop also puts two eigenvalues at $\pm 0.65 = \pm\sqrt{0.43}$.
  - At even lags their powers equal 0.43^ℓ; at odd lags they cancel

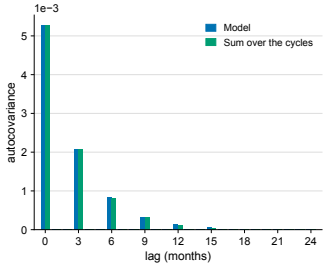
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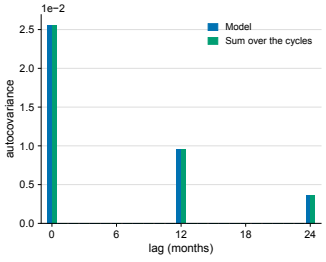
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Lags 0 to 24 months: autocovariances of the model; sum of the cycle contributions



(a) Distilleries: self-loop 0.39, 3 months

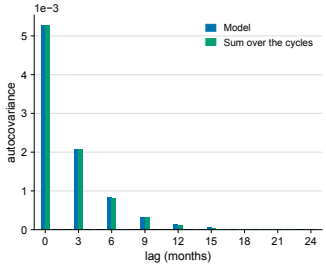


(b) Missile propulsion: self-loop 0.37, 12 months

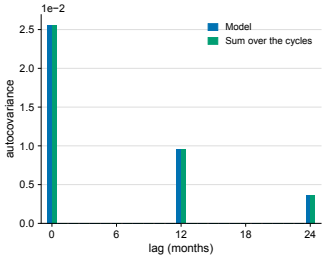
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- All sectors, lags 1 to 24: pooled (weighted) $R^2 = 0.97$, median sector 0.85.

Delays and the Business Cycle

From one sector to the aggregate

- So far, we have studied individual sectors $\Rightarrow$ cycles are key
- What do the delays do to aggregate fluctuations? Rest of network matters

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Setting

- Productivity: independent random walks, one per sector
- Aggregate $\hat{y}_t$ is industrial production: value-added weighted sum of sectors

$$\Delta \hat{y}_t = \sum_n \mu_n \Delta \hat{y}_{nt}$$

- Static network: every shock moves $\hat{y}_t$ at once, growth is white noise

Real GDP and the other aggregates

Aggregate autocovariance: pairs of sectors

- Autocovariance of $\Delta\hat{y}_t$ at lag k: pairs of sectors, m first, then n, k months later

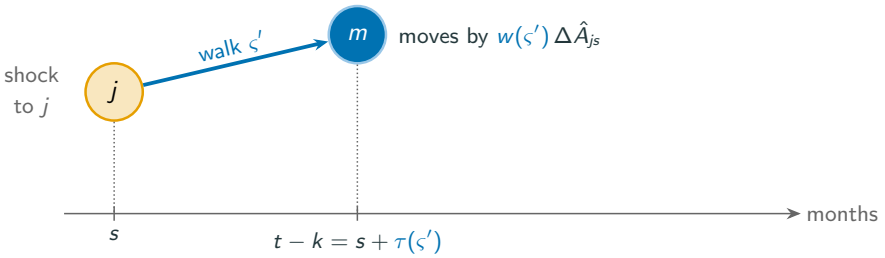
$$\gamma_Y(k) = \sum_n \sum_m \mu_n \mu_m \operatorname{Cov}(\Delta\hat{y}_{nt}, \Delta\hat{y}_{m,t-k})$$

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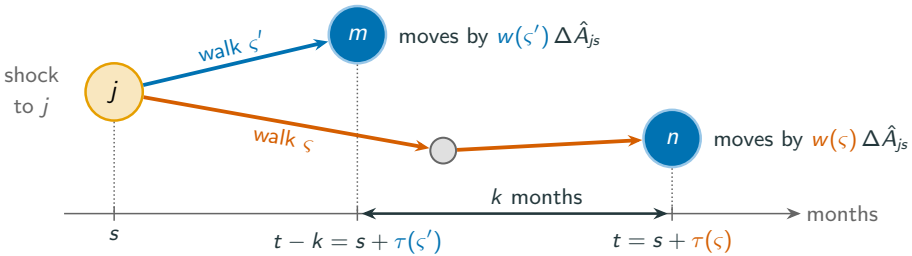
$$\begin{aligned}\gamma_Y(k) &= \sum_n \sum_m \mu_n \mu_m \operatorname{Cov}(\Delta\hat{y}_{nt}, \Delta\hat{y}_{m,t-k}) \\ &= \underbrace{\sum_n \mu_n^2 \operatorname{Cov}(\Delta\hat{y}_{nt}, \Delta\hat{y}_{n,t-k})}_{\text{own terms: last section}} + \underbrace{\sum_{n \neq m} \mu_n \mu_m \operatorname{Cov}(\Delta\hat{y}_{nt}, \Delta\hat{y}_{m,t-k})}_{\text{cross terms: } m \text{ moves, then } n,\ k \text{ months later}}\end{aligned}$$

A cross term: one shock, two walks



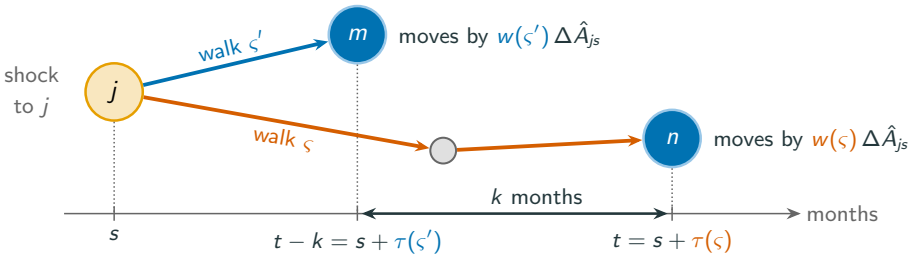
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- A shock to j reaches m along a walk ς'
- It reaches n along a longer walk ς, k months later

A cross term: one shock, two walks



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- It reaches n along a longer walk ς, k months later
- One shock moves both: add up over shocks and pairs of walks k months apart

Formally

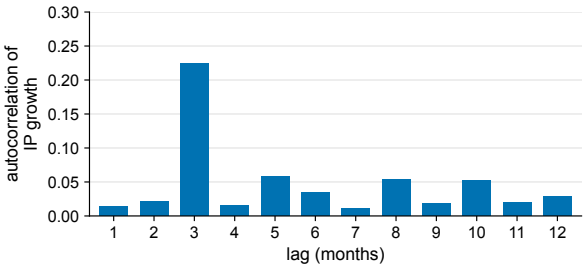
$$\mathrm{Cov}(\Delta\hat{y}_{nt}, \Delta\hat{y}_{m,t-k}) = \sum_j \sigma^2(\hat{A}_j) \sum_{\substack{\varsigma:\, j\to n,\ \varsigma':\, j\to m \\ \tau(\varsigma)-\tau(\varsigma')=k}} w(\varsigma)\, w(\varsigma')$$

Aggregate growth is persistent

- Static: white noise. LP83: gone within a quarter. The measured delays?

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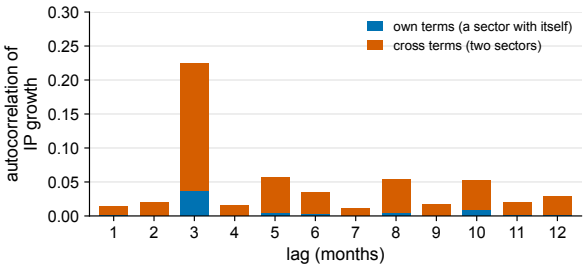
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- Autocorrelation out to a year, where the two alternatives have none

Aggregate growth is persistent

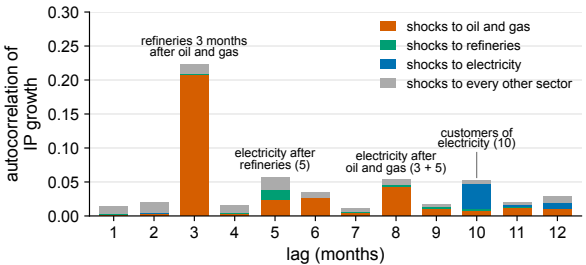
- Static: white noise. LP83: gone within a quarter. The measured delays?



- Cross terms explain most of the aggregate autocorrelation

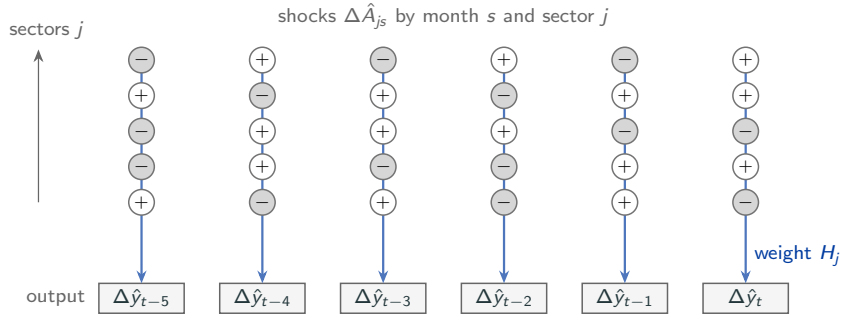
Aggregate growth is persistent

- Static: white noise. LP83: gone within a quarter. The measured delays?



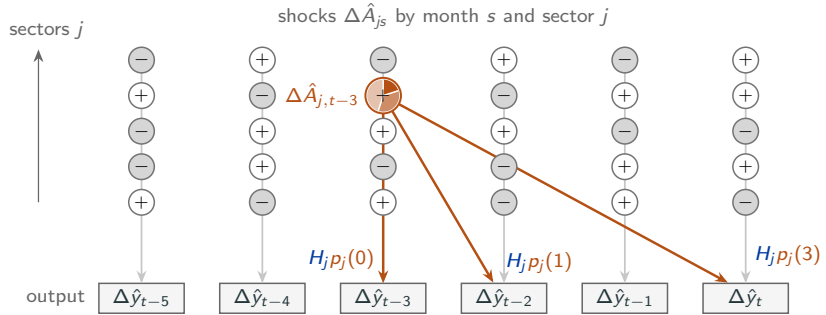
- Energy shapes the persistence of IP:
  - Oil and gas shocks (orange, delay 3): $k = 3$ and $k = 8$ $(3 + 5$, through refineries)
  - Refinery shocks (green) at $k = 5$, electricity shocks (blue) at $k = 10$

Which shocks enter this month's growth



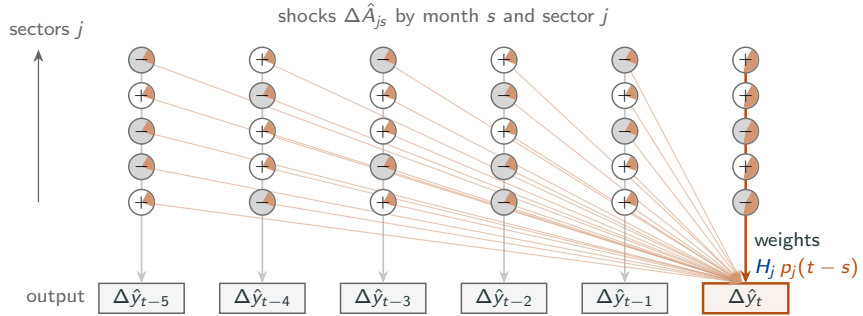
- Static network: output at t adds the sectoral shocks at t, with a total weight of H_j.
  - One averaging: across sectors, where the $+$ and $-$ shocks of the month partly cancel

Which shocks enter this month's growth



- With delays, a shock at s is cut into slices that reach output in different months
  - A share $p_j(d)$ of it arrives d months later

Which shocks enter this month's growth



- Output at t then adds slices of past shocks
  - Two averagings: across sectors and across time, where many more $+$ and $-$ slices cancel

The delays lower the variance of growth at every horizon

Proposition (variance ratio)

With independent random-walk productivities, for every horizon $k \geq 1$,

$$\frac{\operatorname{Var}\left(\hat{y}_t - \hat{y}_{t-k}\right) \text{ with delays}}{\operatorname{Var}\left(\hat{y}_t - \hat{y}_{t-k}\right) \text{ in the static network}} \leq 1$$

The ratio is weakly increasing in k and tends to one as $k \to \infty$.

Walks Two averagings Proof Arrival dates

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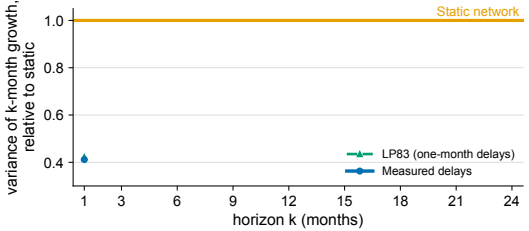
The ratio is weakly increasing in k and tends to one as $k \to \infty$.

- Short horizon: growth collects a slice of each of many past shocks, and they average out
- Long horizon: growth collects almost every slice of each shock, as in the static network

Walks Two averagings Proof Arrival dates

The delays remove a third of the one-year variance

$$\text{variance ratio at } k \text{ months} = \frac{\text{Var}\left(\hat{y}_t - \hat{y}_{t-k}\right) \text{ with delays}}{\text{Var}\left(\hat{y}_t - \hat{y}_{t-k}\right) \text{ in the static network}}$$

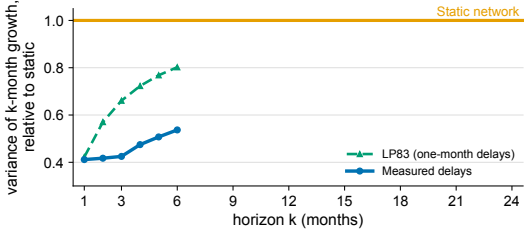


- Static network: output moves right away. With delays: output moves little by little, and each month averages pieces of many past shocks: 0.41 of the static variance (LP83 too)

Whole economy | Common shock | Spectrum

The delays remove a third of the one-year variance

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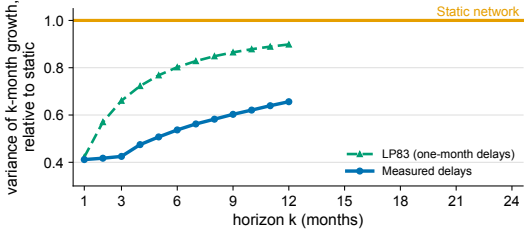


- Half a year: the pieces less than six months apart are back in one window: 0.54

Whole economy | Common shock | Spectrum

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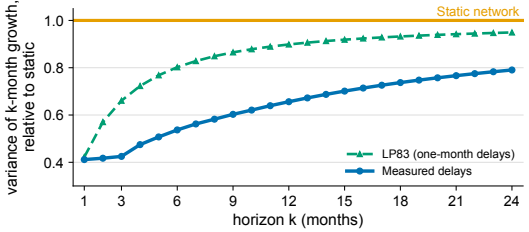


- One year: 0.66. LP83: 0.90, its pieces are at most a few months apart

Whole economy | Common shock | Spectrum

The delays remove a third of the one-year variance

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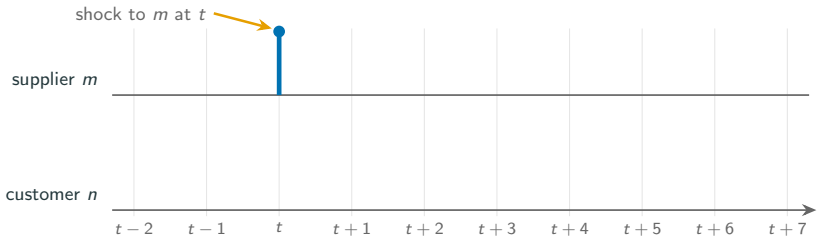


- The long-run variance is the same as in the static model.

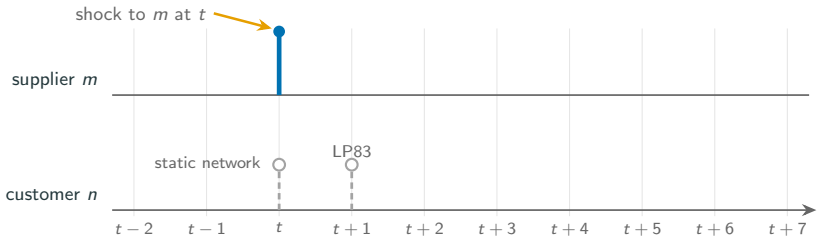
Whole economy | Common shock | Spectrum

Empirical Evidence

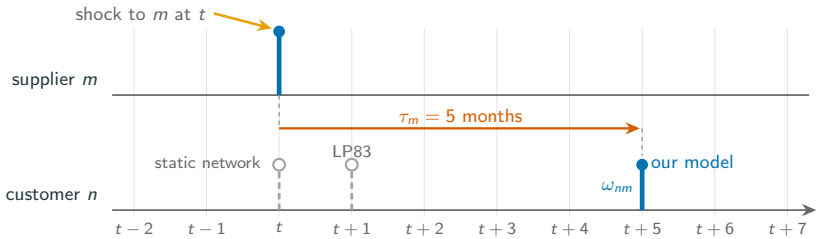
A supplier's shock reaches its customers exactly one delay later



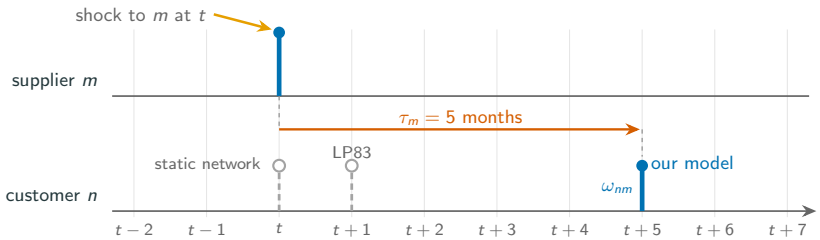
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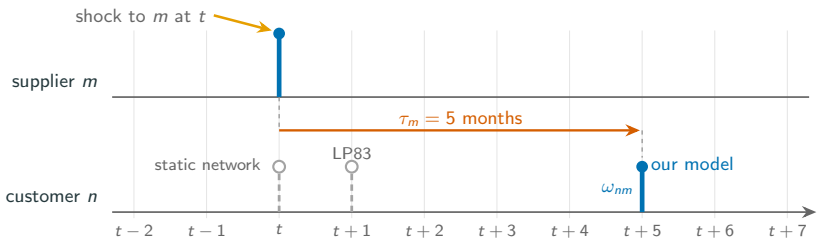
Law of motion

$$\hat{y}_{nt} = \hat{A}_{nt} + \sum_m \omega_{nm}\, \hat{y}_{m,t-\tau_m}$$

In growth rates

$$\Delta\hat{y}_{nt} = \Delta\hat{A}_{nt} + \underbrace{\sum_m \omega_{nm}\, \Delta\hat{y}_{m,t-\tau_m}}_{X_{nt}^{(\tau)},\ \text{the } \tau\text{-matched composite}}$$

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- No free parameter: the shares, the delays and the outputs are all measured

The regression, and what each economy predicts

$$\underbrace{\Delta \hat{y}_{nt} = \Delta \hat{A}_{nt} + X_{nt}^{(\tau)}}_{\text{the model}} \quad \Longrightarrow \quad \underbrace{\Delta \hat{y}_{nt} = \beta_{\tau} X_{nt}^{(\tau)} + \alpha_n + \delta_t + s_{n,\text{mo}(t)} + \varepsilon_{nt}}_{\text{the regression}}$$

fixed effects: industry, month, industry $\times$ calendar month (not seasonally adjusted series)

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Regression coefficient in ideal setting

	Our model	Static network	LP83
β_τ with random-walk productivity	1	0	≈ 0
the customer moves	τ_m months later	at once	one month later

Bringing the model to the data

- Monthly industrial production (Fed G.17)
  - 104 industries matched to the input-output sectors, 1972–2026
  - Manufacturing, mining and utilities: where the delays are best measured

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  2. Some suppliers outside industrial production are missing

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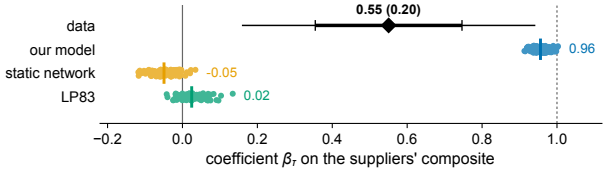
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Proposition

- Solution
  - Estimate AR(1) productivities with a common factor
  - Simulate the full economy (371 sectors) and measure it exactly like the data (dropped sectors, aggregation)
  - Run the same regressions on the data and on 100 simulated panels per economy

Customers move with their suppliers at the measured delays

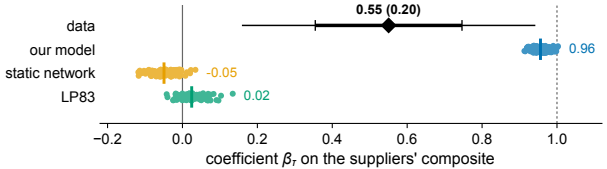


Panel of 104 industries; s.e. clustered by industry. One dot per simulated panel, 100 per economy.

- The data estimate is above every one of the 100 static and 100 LP83 draws

Placebo Robustness Echo

Customers move with their suppliers at the measured delays

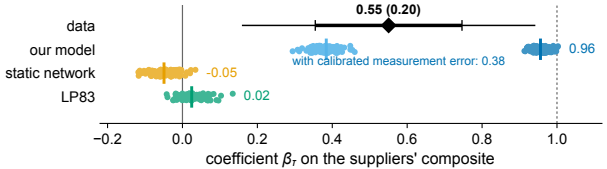


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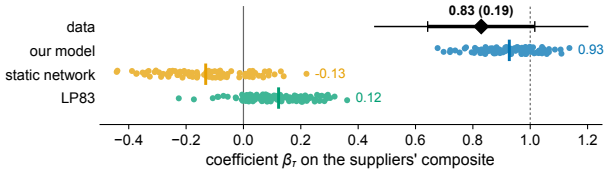
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- The data estimate is above every one of the 100 static and 100 LP83 draws
- Our model has too much action
  - Other mechanisms are obviously at work in reality ⇒ lower coeff.
- Measurement error takes the model to 0.38, within one s.e. of the data

Placebo Robustness Echo

Total industrial production follows its suppliers' delays

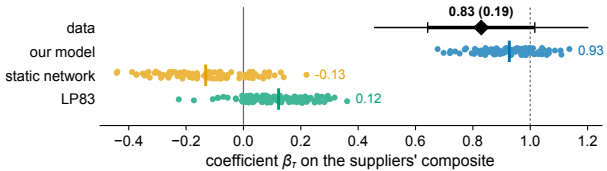
- Same regression on total industrial production and its composite $\bar{X}_t^{(\tau)}$



Total industrial production and $\bar{X}_t^{(\tau)}$: value-added-weighted sums over the 104 industries. Newey–West s.e., twelve lags.

Total industrial production follows its suppliers' delays

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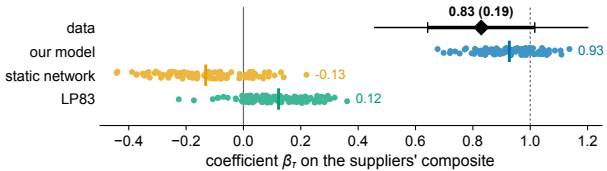


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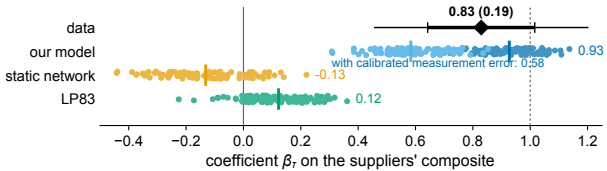


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Total industrial production and $\bar{X}_t^{(\tau)}$: value-added-weighted sums over the 104 industries. Newey–West s.e., twelve lags.

- More than three s.e. above the static and LP83 means
- Our model performs well
  - With calibrated measurement error the model moves to 0.58,

$\Rightarrow$ The mechanism is visible in aggregate industrial production

Robustness

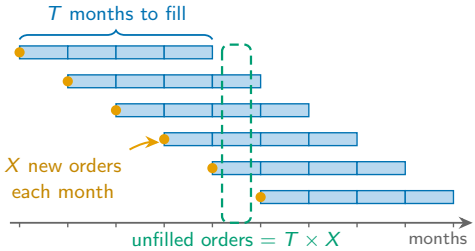
Conclusion

- Delays do not change the total effect of a shock. They spread it over the months that follow, at a welfare cost
- A cycle brings a shock back after one lap, these generate recurrent echoes
- Delays make aggregate growth persistent, and make it autocorrelated even at long lags
- Evidence: in monthly industrial production customers move with their suppliers at the measured delays, and so does total industrial production. A static network or one-month delays cannot replicate these patterns

More in the paper

- Bottlenecks and the welfare cost of delays
- The whole economy, a common shock, and the spectrum of aggregate output
- Delay shocks: disruptions to deliveries

The backlog ratio measures the delay



- Backlog ratio = unfilled orders / monthly shipments = months needed to clear the order book. In the model's steady state it equals the delay exactly
- Measured for 301 of the 371 sectors, 62.6 percent of value added; the rest receive the average of the measured manufacturing sectors, 4 months
- Delays are rounded to whole months and censored to $[1, 24]$

Back

The three sources, and how much they agree

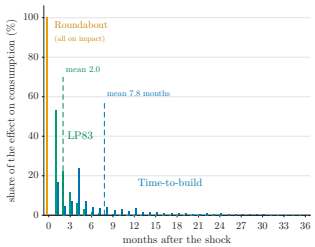
Source	Measure	Coverage	Years
Census M3	unfilled orders / monthly shipments, monthly survey of manufacturers	7 durable-goods industries (NAICS 331–337), one ratio each	1992–2024
Compustat	order backlog / monthly sales, annual filings of listed firms, averaged by sector (3 firms or more)	204 sectors, 112 used where the Census is silent	1992–2024
EU survey	"months of production assured by the current order book", asked of euro-area manufacturers	manufacturing industries (NACE divisions); 65 sectors used	2005–2024

- Same level where two sources overlap: 5.5 vs 5.6 months (Census, Compustat; 92 sectors), 4.7 vs 4.6 (Compustat, EU; 133 sectors)
- Same ranking: the EU survey orders the seven Census industries exactly as the Census does, about a month longer
- Sector by sector, Census and Compustat agree weakly (rank correlation 0.25): one ratio per Census industry, a handful of firms per Compustat sector. Hence the order Census, Compustat, EU

Back

The average duration of a shock

- Shocks are i.i.d. from now on: all persistence comes from propagation
- A shock reaches the household walk by walk. ϕ_t: the share of its total effect ($\mathrm{CIR} = \sum_t \hat{C}_t$) that arrives in month t
- Average duration $\mathcal{T} = \sum_t t\,\phi_t$: the mean arrival date



Unit shock to every sector: ϕ_t by month

- Static network: everything on impact
- LP83: nine tenths within three months
- Measured delays: waves over two years, a fifth still to come after a year. $\mathcal{T} = 7.8$ months vs 2.0
- Single sectors: two fifths have $\mathcal{T} \geq 12$ months; guided missiles 54 (6 in LP83)

Next: delays × traffic ◂ Back

A common four-month delay gives the average, without the slow sectors

- Average duration = passages through production (2.0) $\times$ average delay per passage
- LP83 with every delay at four months, the average measured delay:

	LP83 ($\tau = 1$)	Common delay ($\tau = 4$)	Measured delays
Duration of an aggregate shock (months)	2.0	8.0	7.8
Welfare cost (% of the static value)	0.6	2.6	2.5
One-year variance ratio (industrial production)	0.90	0.66	0.66
Slowest sector (months)	6	24	54
Std. dev. of sectoral durations (months)	1.1	4.4	7.2

- The level of the delays sets the averages; the measured delays set which sectors are slow
- Data: customers follow their suppliers at the measured delays, 0.55 (0.20); at a common four-month lag, -0.11 (0.08)

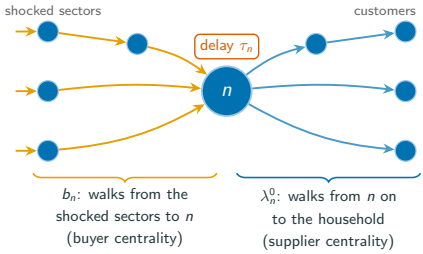
Placebo

Back

Duration = delays × traffic

Proposition (average duration)

$$\mathcal{T} = \sum_n \tau_n \nu_n, \qquad \nu_n = \frac{b_n \times \lambda_n^0}{\mathrm{CIR}}$$



- ν_n: the traffic through n, the expected number of passages
- $b_n = [(\mathbf{I} - \mathbf{\Omega})^{-1}\hat{\mathbf{A}}]_n$: the walks from the shocked sectors into n
- λ_n^0: the walks from n on to the household (static Domar weight)
- No delay in ν_n: the network sets the walks, the delays only the dates. $\mathcal{T}$ is linear in the delays

Back

The welfare cost of delays

- Goods delivered later are worth less. With i.i.d. shocks, the welfare value of a shock is $\Delta V = \sum_{t \geq 0} \beta^t \hat{C}_t$

Proposition (welfare value of a shock)

$$\Delta V = \underbrace{\mathrm{CIR}}_{\text{value in the static network}} \times \underbrace{E\left[\beta^D\right]}_{\text{discount over the wait}} = \mathrm{CIR}\left(1 - r\mathcal{T} + O(r^2)\right)$$

D: the month in which the effect reaches the household, $E[D] = \mathcal{T}$; $r = -\log \beta$

- The delays cost the share $r\mathcal{T}$ of the static value: 2.5% at $r = 4\%$ a year (two thirds of a percent in LP83)
- A shock to guided missiles delivers 16% less welfare than in the static network

Back

Bottlenecks: the sectors whose delays matter the most

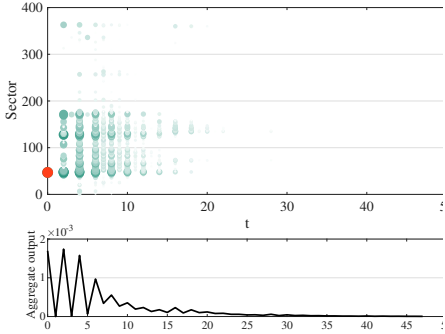
- One more month in n's delay adds its traffic $v_n = b_n \lambda_n^0 / \mathrm{CIR}$ to the duration ($r\, v_n$ to the welfare cost): a bottleneck is central as a buyer (much of the shock reaches it) *and* as a supplier (much of it goes on to the household)

Sector	v_n	b_n	λ_n^0	sold on through customers (%)
Owner-occupied housing	0.079	1.26	0.119	0
Hospitals	0.078	1.81	0.081	2
Other real estate	0.066	2.16	0.058	99
Petroleum refineries	0.051	2.47	0.039	38
Insurance carriers	0.039	1.88	0.040	40
Offices of physicians	0.034	1.69	0.038	0

- Two routes to the top: large sectors selling straight to the household (housing, hospitals), and hubs whose output passes on through others (real estate, refineries). No delay data in the ranking

Back

A shock to nonferrous metal smelting spreads in two-month waves



Shock of 1% to sector #331410 (red dot at $t = 0$). Top: response of each of the 371 sectors by month (dot size and shade = magnitude, log scale). Bottom: aggregate output, in percent.

Back

What if O is not diagonalizable?

- The expansion $\mathbf{Y}_t = \sum_j \xi_j^t \, a_j \mathbf{v}_j$ needs a basis of eigenvectors. Without one, the Jordan form gives

$$\mathbf{Y}_t = \sum_j p_j(t)\, \xi_j^t \, \mathbf{v}_j, \qquad p_j \text{ a polynomial in } t, \text{ a constant when } \xi_j \text{ is not repeated}$$

- Our $\mathbf{O}$ has 8,533 eigenvalues. The 1,475 nonzero ones are distinct, so their terms are exactly ξ_j^t
- The eigenvalue 0 is repeated, so $\mathbf{O}$ is not diagonalizable. Its terms $p(t)\, 0^t$ vanish after τ_{max} months: they are the part of the shock that never returns. Example: $\mathbf{O} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, shock $(0, 1)$: response $(0, 1)$, then $(1, 0)$, then nothing
- A repeated *nonzero* eigenvalue would give $(a + bt)\, \xi^t$: same decay rate, same period. Not our case

$\Rightarrow$ Oscillation and decay live in the nonzero eigenvalues, where the expansion is exact

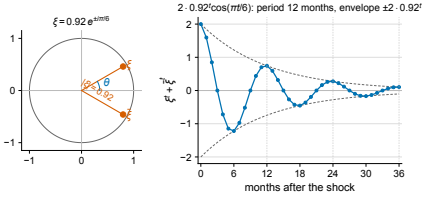
Back

How a pair of complex eigenvalues makes a real oscillation

- **O** is real, so a complex eigenvalue $\xi = |\xi| e^{i\theta}$ comes with its conjugate $\bar{\xi}$, and the two terms add up to a real number:

$$a\,\xi^t + \bar{a}\,\bar{\xi}^t = 2|a|\,|\xi|^t \cos(\theta t + \varphi)$$

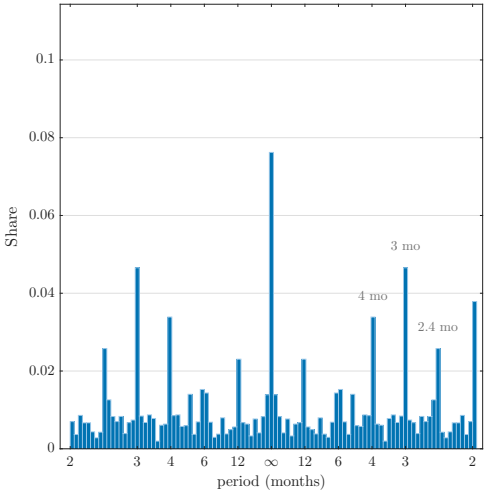
- A damped cosine: period $2\pi/\theta$ months, envelope $|\xi|^t$. The sizes a, $\bar{a}$ only set the amplitude and the phase φ



The annual root of the missile-propulsion self-loop: $|\xi| = 0.37^{1/12} = 0.92$, $\theta = 2\pi/12$

Back

The periods of the eigenvalues

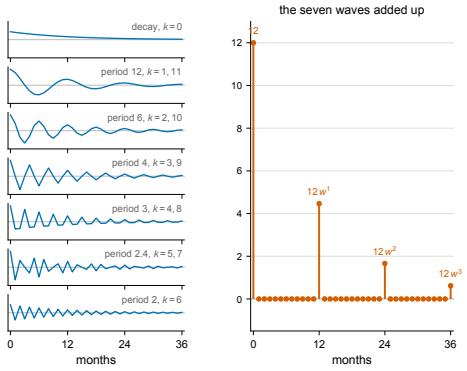


- Each eigenvalue of **O** (measured delays) by its period $2\pi/\theta$ in months, weighted by its modulus
- Center (∞): positive real eigenvalues, which do not oscillate
- Mass at 2, 3, 4 and 12 months, the durations of the heavy self-loops
- A cycle of duration τ also has roots at the periods τ/k, its harmonics: the twelve-month self-loops of transportation equipment also place eigenvalues at 4 months

Back

Why the harmonics: a spike every τ months is the sum of all τ roots

- The τ roots of $\xi^\tau = w$ all have modulus $w^{1/\tau}$: one smooth decay, the fundamental (period τ) and the harmonics (periods $\tau/2$, $\tau/3, \ldots$). Conjugate roots pair into real waves
- Why all of them are there: the law $\hat{y}_t = w\,\hat{y}_{t-\tau}$ fixes nothing within a lap, so any shape of length τ repeats, scaled by w per lap
- A spike every τ months is the sum of every wave with equal weight; between the spikes they cancel



$\tau = 12$, $w = 0.37$ (missile propulsion). Left: the decay and the six waves (periods 12, 6, 4, 3, 2.4, 2 months). Right: their sum, a spike of $12\,w^m$ every 12 months

Back

Why every solution of $R_n(\xi^{-1}) = 1$ is an eigenvalue

- After the shock, n moves at date t only through a walk from n back to n. Cut it at its first return: a first-return walk ς (weight $w(\varsigma)$, duration $\tau(\varsigma)$), then any walk from n to n over the remaining $t - \tau(\varsigma)$ months. Hence

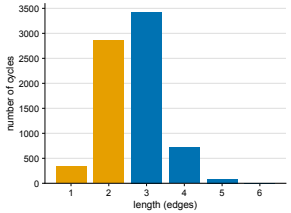
$$\hat{y}_{n,t} = \sum_{\varsigma} w(\varsigma)\, \hat{y}_{n,t-\tau(\varsigma)} \qquad (t > 0)$$

- ξ^t solves this linear equation iff $\sum_\varsigma w(\varsigma)\xi^{-\tau(\varsigma)} = 1$, that is $R_n(\xi^{-1}) = 1$
- Each such ξ is an eigenvalue of $\mathbf{O}$ (paper): the generating function of the walks from n back to n is $1/(1 - R_n(z))$, and its poles are the inverse eigenvalues
- One cycle through n: $R_n(z) = wz^\tau$, so $\xi^\tau = w$. No cycle anywhere: $R_n = 0$ for every n, and $\mathbf{O}$ is nilpotent
- U.S. network: 345 of 371 sectors share cycles; only six self-loops are exactly isolated. Exact roots are rare, close ones are not (main slides)

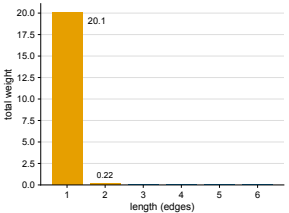
Back

The cycles of the U.S. network: a census

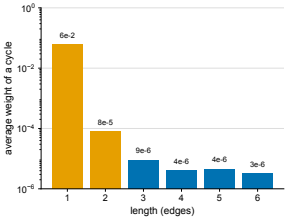
Every cycle of the network with weight above 10^{-6}: 7,414 cycles, many of them long



(a) Number of cycles



(b) Total weight

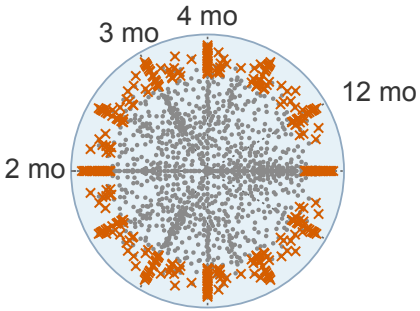


(c) Average weight of a cycle

- Almost all the weight is on the self-loops: 97% of the weight that returns to a sector; each longer cycle is tiny (c)
- Largest self-loops: nonferrous metal 0.43, beef cattle 0.41, distilleries 0.39; heaviest two-sector cycle 0.025. A self-loop above 0.1 in one sector in six (13% of output)

Back

The self-loops predict the eigenvalues of the U.S. network



- Dots: the eigenvalues of $\mathbf{O}$. Crosses: the roots $\xi^{\tau_n} = \omega_{nn}$ of every self-loop (modulus above 0.7)
- The slowest-dying eigenvalues (modulus above 0.85, 102 of them): each one within 0.02 of a self-loop root, and each root (8 sectors) within 0.02 of an eigenvalue
- Modulus above 0.7: 88% of the 415 eigenvalues are within 0.02 of a self-loop root
- Why: a heavy self-loop dominates R_n, so its roots $\xi^{\tau_n} = \omega_{nn}$ nearly solve $R_n(\xi^{-1}) = 1$

Back

Every pair of walks from one shock: propagation or separate routes

- $\gamma_Y(k)$ adds up pairs of walks from one shock, the earlier ending at m, the later at n, k months apart. Either the later walk continues the earlier one (propagation) or the two part before m (separate routes)



(a) propagation: the shock moves m, then travels on to n (b) separate routes: the shock reaches m and n by different walks

Proposition (the aggregate autocovariances by pairs of walks)

$$\gamma_Y(k) = \underbrace{\sum_m q_m \mu_m h_m(k)}_{\text{propagation}} + \underbrace{T(k)}_{\text{separate routes} \geq 0} \qquad (k \geq 1)$$

q_m: the variance of m's output carried by one walk at a time, $\sum_j \sigma_j^2 \sum_{\text{walks } j \to m} \text{weight}^2$; $h_m(k)$: the response of the index to m after k months

- Shares of $\sum_{k=1}^{12} \gamma_Y(k)$, industrial production: propagation between two sectors 68% (one link 45, chains 23), $n = m$ laps 10%, $n = m$ two routes 2%, separate routes 20%

Arrival dates

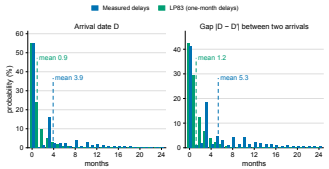
Back

The arrival-date view: the delays enter through one distribution

Proposition (aggregate autocovariances)

Draw a walk from j at random (weight $\times$ index weight of its last sector); its arrival date D_j is when it reaches the index. With independent random-walk productivities and i.i.d. draws D_j, D_j',

$$\gamma_Y(k) = \sum_j \pi_j \Pr\left(D_j' - D_j = k\right), \qquad \pi_j = \sigma_j^2 H_j^2, \quad H_j = \sum_d h_j(d)$$



Industrial production: D (left), the gap $|D - D'|$ (right)

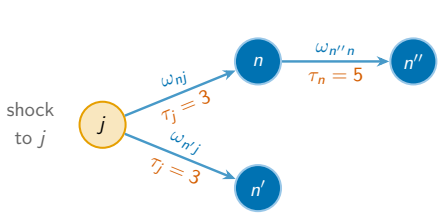
- π_j: variance $\times$ long-run effect squared, free of the delays
- The delays enter only through the gaps $|D' - D|$ (5.3 months on average, 1.2 in LP83); at $k = 0$, the variance of growth relative to the static network is the probability that two arrivals coincide
- Variance ratio at horizon k: $1 - E[\min\{|D - D'|, k\}]/k$

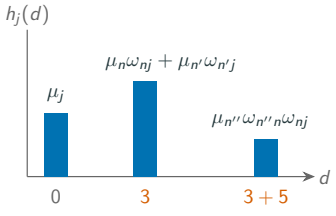
Pairs of walks

Back

Aggregate growth adds up past shocks along walks

- A shock to j travels every walk ς out of j to move ς's end sector n after its duration $\tau(\varsigma)$





- Let $h_j(d)$ be the response of agg. output d months after a shock to j
- Aggregate growth adds up the past shocks of every sector

Formally

$$\Delta \hat{y}_t = \underbrace{\sum_{j}}_{\text{sectors}} \underbrace{\sum_{d \geq 0}}_{\text{arrival dates}} h_j(d)\, \Delta \hat{A}_{j,t-d}$$

Back

Aggregate growth, walk by walk

- Sector n moves with every shock that reaches it along a walk; aggregate growth weights the sectors by μ_n

$$\Delta \hat{y}_t = \sum_n \mu_n \, \Delta \hat{y}_{nt} = \sum_j \sum_n \mu_n \sum_{\varsigma : j \to n} w(\varsigma) \, \Delta \hat{A}_{j,t-\tau(\varsigma)}$$

- Collect the walks out of j by their duration d

$$h_j(d) = \sum_n \mu_n \sum_{\varsigma : j \to n,\ \tau(\varsigma) = d} w(\varsigma) \qquad \Rightarrow \qquad \Delta \hat{y}_t = \sum_j \sum_{d \geq 0} h_j(d) \, \Delta \hat{A}_{j,t-d}$$

- The total $H_j = \sum_d h_j(d)$ is the response to the same shock in the static network: the delays change the dates, not the total

Back

Two averagings: across sectors and across time

- Split the response $h_j(d)$ of output to a shock to j into size and timing

$$h_j(d) = \underbrace{H_j}_{\substack{\text{size: the long-run response } \sum_d h_j(d), \\ \text{the static network's Hulten weight}}} \times \underbrace{p_j(d)}_{\substack{\text{timing: the share } h_j(d)/H_j \\ \text{arriving } d \text{ months later}}}$$

- Shocks independent across sectors and months, so the variance adds up shock by shock

$$\mathrm{Var}(\Delta \hat{y}_t) = \sum_j \underbrace{\sigma^2(\hat{A}_j)\, H_j^2}_{\substack{\text{the static formula: Hulten, granularity}}} \underbrace{\sum_d p_j(d)^2}_{\substack{\text{new: one in the static network}}}$$

- Many small shares average out, as many small Hulten weights do. The static network averages across sectors only ($p_j(0) = 1$); the delays add averaging across time

Back

The variance ratio: proof

- Growth over k months adds the slices that arrive in the window (a: months since the shock)

$$\hat{y}_t - \hat{y}_{t-k} = \sum_j \sum_{a \geq 0} H_j \Big(\sum_{d=a-k+1}^{a} p_j(d) \Big) \Delta \hat{A}_{j,t-a}$$

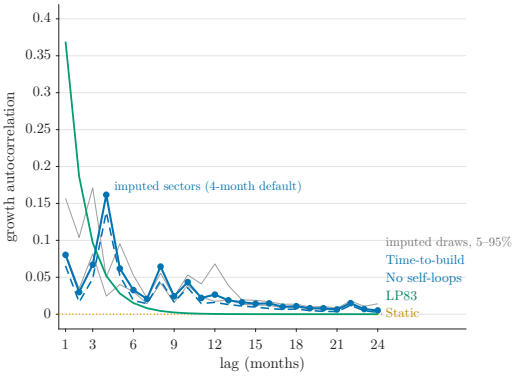
- Shocks are independent across sectors and months:
  $\mathrm{Var}\left(\hat{y}_t - \hat{y}_{t-k}\right) = \sum_j \sigma^2(\hat{A}_j) H_j^2 \sum_{a \geq 0} \left(\sum_{d=a-k+1}^{a} p_j(d) \right)^2$
- Slices d and d' are in the same window for $(k - |d - d'|)^+$ values of a; D, D' i.i.d. from p_j

$$\sum_{a \geq 0} \Big(\sum_{d=a-k+1}^{a} p_j(d) \Big)^2 = \sum_{d,d'} p_j(d)\, p_j(d')\, (k - |d - d'|)^+ = k - E_j \left[\min \left\{ |D - D'|, k \right\} \right]$$

- Static network ($p_j(0) = 1$): variance $k \sum_j \sigma^2(\hat{A}_j) H_j^2$. Divide: the ratio is $1 - E\left[\min\{|D - D'|, k\}\right]/k$, where E averages the E_j with these weights
- $p_j(d) \geq 0$ (walk weights): $\min\{|D - D'|, k\}/k$ is in $[0, 1]$, weakly falls as k rises, and tends to zero

Back

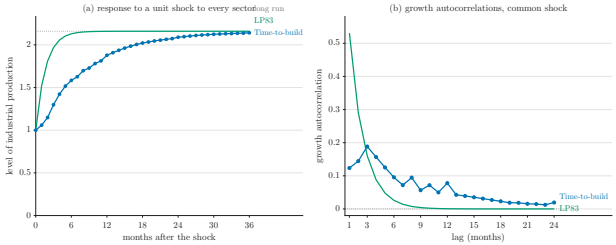
The whole economy: the peak is the imputed delay, the variance ratio is not



- Output index over all 371 sectors: the 70 sectors without backlog data (services) carry 46% of the shock weights, against 4% in industrial production
- Peak at 4 months (0.16), the imputed default: 87% of it comes from these sectors; set the imputed delay to k and the peak moves to k
- Draw the imputed delays at random from the measured ones (band): the peak disappears (0.04)
- The one-year variance ratio is 0.65 under the default and 0.66 across the draws: it depends on the delays only through the gaps

Back

A common shock: the delays remove even more variance

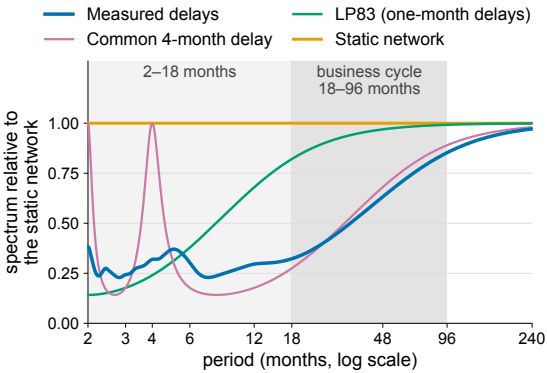


- A unit shock to every sector at once (the innovation of a common random walk), industrial production. Left: the level keeps rising for months after impact, mean arrival date 5.0 months (1.2 in LP83). Right: growth positively autocorrelated at every short lag, the oil-and-gas peak on top
- The delays remove almost half of the annual variance (ratio 0.55), more than for idiosyncratic shocks: the arrivals of a common shock are the most spread out (mean gap 7.1 months against 5.3)

Back

The same damping in the frequency domain, with persistent productivity

- Spectrum of the index under the estimated AR(1) productivities with a common factor (closed form in the paper): same network, same shocks, only the delays differ



Spectrum relative to the static economy, by period

- Equal at the longest periods: the long-run multiplier is untouched
- Business-cycle band (18 to 96 months): 0.64 of the static level under the measured delays; total variance 0.82
- One common delay for every sector: sharp peaks at its period and harmonics; the dispersion of the measured delays disperses them

Back

With persistent productivity the ideal coefficients move a little

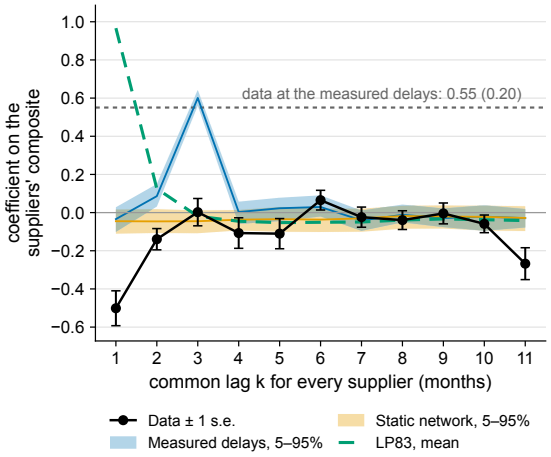
Proposition

Setup: *independent AR(1) productivities with persistence* $\rho_n \in [0, 1)$*; every delay at least one month.*
Our model: $\beta_\tau = 1 + Cov(\Delta \hat{A}_{nt}, X_{nt}^{(\tau)})/Var(X_{nt}^{(\tau)}) \leq 1$*, the gap of order* $(1 - \rho_n)^2$ *times the weight of the loops through* n*.*
Static network: $\beta_\tau \leq 0$.
LP83: *the slope of a proxy (*$X^{(\tau)}$ *for the one-month composite* $X^{(1)}$*), plus the same nonpositive term.*

- Below one in our model: past growth returns through the loops into the composite, negatively correlated with today's growth
- At most zero in the static network: past productivity growth predicts slightly negative values of today's
- Why growth rates: in levels, persistence alone gives 1.5 (static) against 1.7 (ours)
- TFP growth is close to white noise (-0.03 annual, 78 KLEMS series): the gaps are small

Back

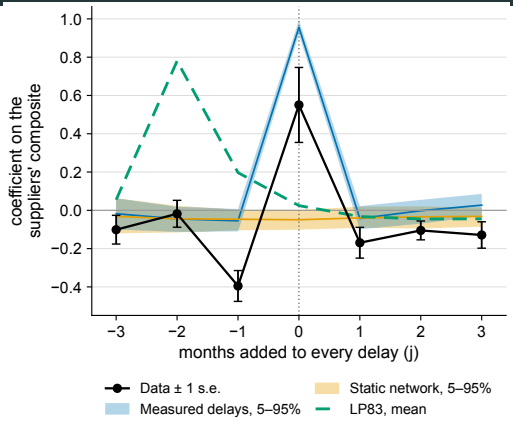
Placebo: a common lag for every supplier carries no signal



- Rebuild the composite with every delay set to the *same* lag k; data with ± 1 s.e., model bands
- Data: at or below zero at every common lag short of a year. The comovement sits only at the measured, heterogeneous delays
- $k = 1$ is LP83's own regression: 0.97 on its panels, -0.50 (0.09) in the data
- $k = 0$ is not identified: a contemporaneous composite contains the industry's own output through the loops of the network

Back

The comovement sits at the measured delays



- Offsets: the data coefficient peaks at the measured delays ($j = 0$) and is at or below zero at every other offset
- Reassign the delays at random across sectors, 500 times: the true assignment beats all 500
- Add as a control the composite rebuilt two and three months away from the delays (common demand): the coefficient is unchanged

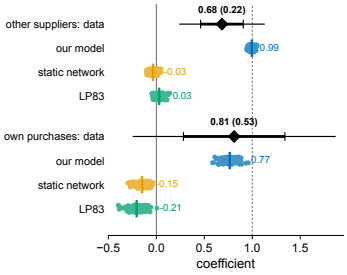
Every delay shifted by a common offset of j months

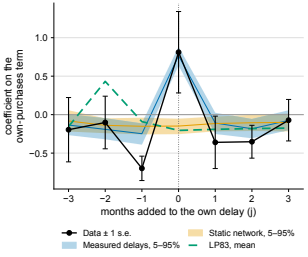
$\Rightarrow$ Shifting the delays by one month removes the comovement; no reassignment reproduces it

Back

The echo in the data: an industry moves with its own past at its own delay

- Self-loops could be an artifact of aggregation. Test: split the composite into the purchases from other suppliers and the industry's own purchases $\omega_{nn}\Delta\hat{y}_{n,t-\tau_n}$ (delays of three months or more). The model predicts a coefficient of one on both





Own term sampled at the delay plus j months

⇒ The echo sits at the measured delay: a peak at $j = 0$, at or below zero elsewhere

Delays of three months or more

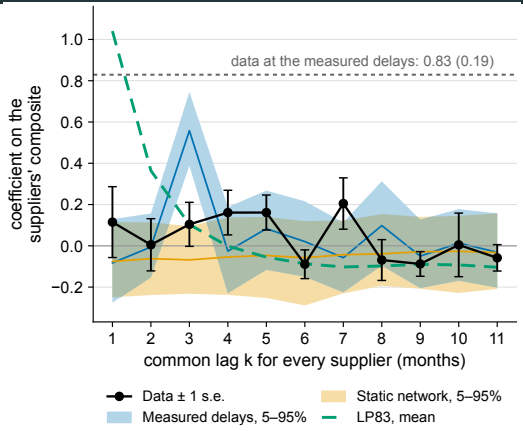
Back

Measurement error and the industries with short delays

- Error in the *level* of a series makes consecutive growth rates negatively correlated: an error that raises one month's level raises that month's growth and lowers the next month's by the same amount
- For an industry with a one-month delay, the own term is its growth one month back, which shares the error of the left-hand side; this pushes its coefficient down
- With a single own term for all industries: 0.24 (0.31) in the data, against 0.75 in our model, -0.11 static, 0.00 LP83: no economy is clearly rejected
- Split by delay: three months or more, 0.81 (0.53), free of the shared error; shorter delays, -0.20 (0.08), the sign the error predicts
- The simulated economies carry no measurement error, so only the first group is comparable with them

Back

The aggregate result: robustness



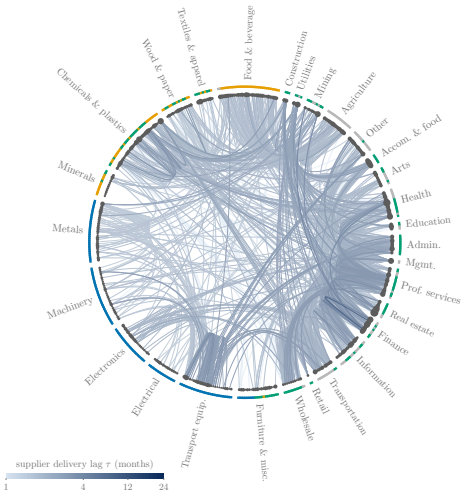
Aggregate regression with every delay set to a common lag k

- Common lags: smaller at every k. At three months, where the model's aggregate growth peaks, the data give 0.10 (0.11) against 0.56: the model's three-month peak is not in the data
- Reassigning the delays at random lowers the coefficient
- Momentum in aggregate growth does not drive the estimate
- Measurement error does not rescue the static network or LP83

Back

Backup

The network and its delays



- Each point is a sector, sized by its Domar weight (its sales over GDP)
- Chords: the 1,500 largest input flows, darker when the supplier's delay is longer
- Outer ring: the source of each sector's delay, colored as in the histogram
- Long delays: durable manufacturing, and project services (engineering, research, computer-systems design)
- Both ship their output throughout the economy

Back

Prices: two markets, one at each end of production

- Prices are the planner's shadow values, in units of utility
- Spot market, delivery today:

$$p_{nt} = \frac{\partial V}{\partial X_{n0,t}} = \frac{\zeta_n}{X_{n0,t}}$$ one more unit of n delivered today

- Forward market, delivery τ_n periods later:

$$q_{nt} = \beta E_t \left[\frac{\partial V}{\partial X_{n,\tau_n-1,\,t+1}} \right] = \beta^{\tau_n} \frac{\zeta_n}{y_{nt}}$$ today's value of that delivery; in steady state $q_n = \beta^{\tau_n} p_n$

- Nominal magnitudes are constant: $p_{nt} X_{n0,t} = \zeta_n$, $q_{nt} y_{nt} = \beta^{\tau_n} \zeta_n$, $\sum_n p_{nt} c_{nt} = 1$

$\Rightarrow$ Forward revenue $\beta^{\tau_n} \zeta_n$ is the marginal social value of a shock to n

Next: value added and GDP ◂ Back

Value added, nominal GDP and Domar weights

- Value is created at production, when the inputs are used (goods in process count as output): the national-accounts convention
- Value added = forward revenue − inputs at spot prices = the wage bill:

$$\mathrm{VA}_n = q_{nt} y_{nt} - \sum_m p_{mt} x_{nm,t} = \alpha_n \beta^{\tau_n} \zeta_n$$

- Nominal GDP $= \sum_n \mathrm{VA}_n$, a constant
- Domar weight = forward sales over nominal GDP. The delays enter only through β^{τ_n}:

$$\lambda_n = \frac{\beta^{\tau_n} \zeta_n}{\mathrm{GDP}^{\mathrm{nom}}} \xrightarrow{\beta \to 1} \lambda_n^0 = \left[(\mathbf{I} - \mathbf{\Omega}')^{-1} \boldsymbol{\gamma} \right]_n, \quad \text{the static network's}$$

The largest Domar weights | Next: real GDP and Hulten | ◂ Back

Real GDP and Hulten's theorem

- Real GDP: the Divisia index of double-deflated value added, $d \log Y_t = \sum_n \frac{\text{VA}_n}{\text{GDP}^{\text{nom}}} d \log Q_{nt}^{\text{VA}}$

$$d \log Q_{nt}^{\text{VA}} = \frac{q_{nt} y_{nt}}{\text{VA}_n} d \log y_{nt} - \sum_m \frac{p_{mt} x_{nm,t}}{\text{VA}_n} d \log x_{nm,t}$$

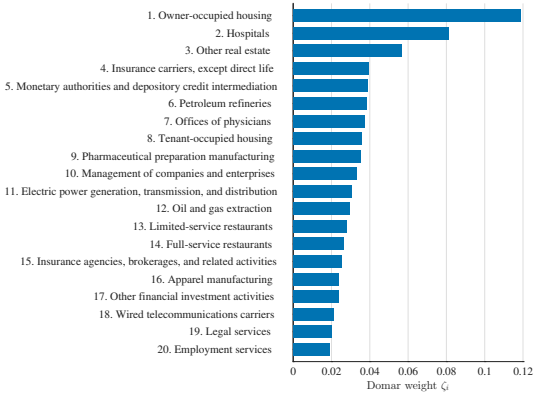
Proposition

Hulten's theorem survives heterogeneous time-to-build: $d \log Y_t = \sum_n \lambda_n d \log A_{nt}$

- With i.i.d. shocks it also measures welfare: $dV = \text{GDP}^{\text{nom}} \times d \log Y_t$
- Real GDP is blind to the delays: output is counted when produced, so with i.i.d. shocks it is white noise

Aggregates ◂ Back

Domar Weights



Top 20 sectors by ζ_n, the value of a delivered good (spot sales; the axis label). The model's Domar weight discounts it for the wait: $\lambda_n = \beta^{\tau_n}\zeta_n/\text{GDP}^{\text{nom}}$.

Prices and GDP

Back

Three aggregates see a shock at different times

Output is $\hat{\mathbf{y}}_t = \mathbf{M}(L)\hat{\mathbf{A}}_t$ with $\mathbf{M}(z) = (\mathbf{I} - \boldsymbol{\Omega}\,\mathbf{diag}(z^{\boldsymbol{\tau}}))^{-1}$, the sum of $w(\varsigma)z^{\tau(\varsigma)}$ over walks

- Real GDP: $d\log Y_t = \sum_n \lambda_n d\log A_{nt}$. White noise with i.i.d. shocks: blind to the delays
- Consumption: $\hat{C}_t = \boldsymbol{\gamma}'\mathbf{diag}(L^{\boldsymbol{\tau}})\mathbf{M}(L)\hat{\mathbf{A}}_t$, one more delivery leg: nothing on impact; its discounted sum is the welfare value of the shock
- Aggregate output: $\hat{y}_t = \boldsymbol{\mu}'\mathbf{M}(L)\hat{\mathbf{A}}_t$, $\mu_n = \mathrm{VA}_n/\mathrm{GDP}^{\mathrm{nom}}$: what industrial production measures, with the full dynamics

$\Rightarrow$ In the static network all three move on impact; the delays separate their timing

◂ Back: Solution ◂ Back: Aggregation

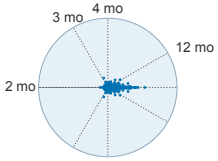
Further related literature

- **Growth accounting with intermediate inputs**: Hulten (1978)
- **Distortions in production networks**: Jones (2011), Liu (2019), Baqaee and Farhi (2019)
- **Delays in supply chains**: Meier (2020)

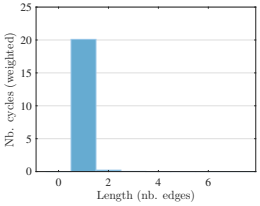
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From Network Cycles to Business Cycles (LP83)

Eigenvalues, LP83



Cycles by duration in LP83 (weighted)



- Every delay is one month, so a cycle's duration is its number of links; almost all the weight is at duration 1, the self-loops
- A one-month self-loop has one root, $\xi = \omega_{nn}$: real and positive, it does not oscillate
- The companion matrix is $\mathbf{\Omega}$ itself; its eigenvalues that are not positive real are small (Liu and Tsyvinski)

$\Rightarrow$ LP83 puts its cycles at a duration of one month, so it barely oscillates

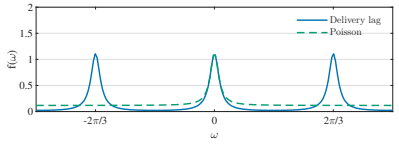
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Poisson Model

- A common way to model delays is Poisson arrival: for a delay τ, delivery with probability $1/\tau$ each period
- Example: sector n with a self-loop of weight w. Autocovariance of its output at lag $k \geq 1$:

$$\gamma_k = w \left(1 - \frac{1}{\tau}\right)^{k-1} \frac{1}{\tau} \sigma^2 (\hat{y}_{nt}) + \text{further laps}$$

With a delivery lag instead: $\gamma_k = w^j \sigma^2 (\hat{y}_{nt})$ at $k = j\tau$, zero at every other lag



Spectrum of a self-loop with $\tau = 3$ and $w = 0.75$: delivery lag vs. Poisson arrival

$\Rightarrow$ Poisson arrival spreads each lap over all later months: the peaks at $2\pi/\tau$ disappear

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